

INTRUSION DETECTION IN WIFI 7 IOT NETWORKS USING EDGE AI

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DECLARATION

I hereby declare that the work have been done by myself and no portion of the work contained in this thesis has been submitted in support of any application for any other degree or qualification of this or any other university or institute of learning.



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ABSTRACT

This project aims to develop an advanced intrusion detection system (IDS) for WiFi 7 IoT networks by leveraging Edge AI capabilities. The system utilizes machine learning algorithms deployed at the network edge to detect and mitigate security threats in real time, protecting IoT devices from increasingly sophisticated attacks. By combining WiFi 7's enhanced network features with the computational efficiency of Edge AI, the proposed solution provides robust security while maintaining low latency and high efficiency for critical IoT applications.

The rapid growth of IoT devices and the adoption of next-generation wireless technologies have significantly expanded the attack surface of modern networks. Traditional centralized intrusion detection approaches often suffer from high latency, scalability limitations, and heavy reliance on cloud infrastructure, making them less suitable for real-time threat detection in resource-constrained IoT environments. To address these challenges, this project adopts a lightweight, machine learning-based anomaly detection approach, such as an autoencoder, which is capable of identifying abnormal network traffic patterns without requiring extensive labelled datasets.

The proposed Edge AI-based IDS is evaluated based on detection accuracy, false positive rate, and processing latency to assess its effectiveness and practicality in WiFi 7 IoT scenarios. The findings demonstrate that deploying intrusion detection mechanisms at the network edge can significantly enhance security performance while reducing response time and network overhead, thereby offering a scalable and efficient solution for securing future IoT networks.

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LIST OF ABBREVIATIONS/ SYMBOLS

AI	Artificial Intelligence
IDS	Intrusion Detection System
IoT	Internet of Things
CNN	Convolutional Neural Network
DDoS	Distributed Denial of Service
FPR	False Positive Rate
ML	Machine Learning
MLO	Multi-Link Operation
Ns-3	Network Simulator 3
PHY	Physical Layer
TCP	Transmission Control Protocol
UDP	User Datagram Protocol

μ	Micro
<i>ms</i>	Miliseconds
<i>mJ</i>	Milijoules

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CHAPTER 1

INTRODUCTION

1.1 Overview

The rapid expansion of Internet of Things (IoT) technologies has led to the deployment of a large number of interconnected devices across various domains such as smart homes, healthcare systems, industrial automation, and smart cities. With the introduction of WiFi 7 (IEEE 802.11be), IoT networks are expected to benefit from significantly higher data rates, ultra-low latency, and improved reliability. These enhancements make WiFi 7 well suited for supporting data-intensive and time-critical IoT applications. However, the increased connectivity and complexity of IoT environments also introduce new security challenges.

IoT devices are often resource-constrained and operate in heterogeneous and distributed network environments, making them vulnerable to a wide range of cyber threats, including denial-of-service attacks, unauthorized access, and malicious traffic injections. Traditional intrusion detection systems (IDS) are commonly centralized and cloud-based, which can lead to high communication overhead, delayed threat response, and scalability issues. Such limitations reduce their effectiveness in next-generation IoT networks that require real-time and low-latency security solutions.

Edge Artificial Intelligence (Edge AI) has emerged as a promising approach to address these challenges by enabling data processing and intelligent decision-making at the network edge, closer to the IoT devices. By deploying machine learning models at edge nodes, intrusion detection can be performed in real time with reduced latency and minimal reliance on cloud infrastructure. This approach not only improves detection speed but also enhances privacy and network efficiency.

This project focuses on the development of an advanced intrusion detection system for WiFi 7 IoT networks using Edge AI. The proposed system utilizes machine learning-based anomaly detection techniques to identify suspicious network

behaviour and potential security threats. By integrating WiFi 7's enhanced networking capabilities with Edge AI's computational efficiency, the system aims to provide robust, scalable, and low-latency security protection suitable for modern IoT environments.

1.1.1 Knowledge Gap

A review of existing literature on intrusion detection systems, Edge AI, and IoT security reveals several important gaps that remain unresolved, particularly in the context of emerging WiFi 7 networks. Most current studies focus on legacy wireless standards such as WiFi 4, WiFi 5, and WiFi 6, and primarily address generic IoT attacks or vulnerabilities related to WPA2 and WPA3 security mechanisms. Although WiFi 7 introduces advanced technologies such as Multi-Link Operation (MLO), wider channel bandwidths up to 320 MHz, and operation within the 6 GHz band, these features are rarely examined from a security perspective. Consequently, there is a lack of WiFi 7 specific threat modeling and attack characterization, leaving new vulnerabilities such as MLO-based attacks and high-band jamming largely unexplored in the literature.

In addition, while Edge AI has been widely proposed as a solution for real-time intrusion detection in IoT environments, most existing studies evaluate their models using offline datasets or simulated settings. These evaluations often emphasize detection accuracy while overlooking critical system-level performance metrics such as inference latency, throughput impact, and energy consumption. This presents a significant limitation, as WiFi 7 is designed to support ultra-low latency applications that require response times below a few milliseconds. The absence of real-time performance validation on edge hardware raises concerns about the practical feasibility of deploying AI-based intrusion detection systems in high-throughput WiFi 7 networks.

Federated learning has also gained attention as a privacy-preserving approach for distributed intrusion detection. However, current implementations assume relatively stable and low-bandwidth communication links between participating nodes. Such assumptions are not well aligned with the dynamic and high-speed nature of WiFi

7, particularly due to its use of Multi-Link Operation across multiple frequency bands. Existing studies do not sufficiently address the challenges of model aggregation latency, synchronization, and communication overhead in WiFi 7 environments. As a result, the integration of federated learning with WiFi 7-based intrusion detection remains an open research problem.

Another notable gap lies in feature engineering for intrusion detection. Most existing systems rely heavily on network-layer and transport-layer features, such as IP addresses, ports, and traffic statistics. Although some research considers link-layer features, these are generally limited to older IEEE 802.11 standards. There is currently little to no work that explores the use of WiFi 7 physical model and MAC layer features, such as MLO-related headers, high-order modulation indicators, or wide-channel utilization patterns. This limits the ability of intrusion detection systems to detect attacks that exploit the unique characteristics of WiFi 7 at the lower layers of the network stack.

Furthermore, many proposed intrusion detection solutions are evaluated using simulated environments or generic IoT devices that do not support WiFi 7. The lack of experimental validation using WiFi 7 ready hardware, such as modern access points or network interface cards, makes it difficult to assess the real-world impact of deploying an intrusion detection system in such environments. In particular, the effects on packet processing capability, network throughput, and latency remain largely unexamined, highlighting the need for realistic testbed-based evaluation.

Robustness against attack, or adversarial robustness, is another area that has received limited attention in current research. While some studies acknowledge the existence of adversarial attacks against machine learning-based intrusion detection systems, few propose concrete defense mechanisms, especially for models deployed at the network edge. In high-speed WiFi 7 environments, attackers may exploit rapid transmission rates to evade detection or poison learning models. The absence of robust defense strategies for Edge AI-based intrusion detection systems presents a critical vulnerability that has yet to be adequately addressed.

Finally, the literature lacks a standardized evaluation framework specifically tailored to WiFi 7 intrusion detection systems. Existing studies employ inconsistent evaluation metrics, often prioritizing detection accuracy without considering latency, reliability, and throughput requirements unique to WiFi 7. Without a unified evaluation framework that combines security effectiveness with system performance constraints, it is challenging to objectively compare existing solutions or determine their suitability for next-generation wireless networks.

Overall, these gaps demonstrate that current intrusion detection research does not sufficiently address the security and performance challenges introduced by WiFi 7, particularly in IoT and edge computing contexts. Addressing these limitations provides a strong motivation for this project, which aims to design and evaluate an Edge AI based intrusion detection system specifically tailored for WiFi 7 IoT networks. lines.

1.2 Problem Statement

The rapid growth of Internet of Things (IoT) applications, combined with the introduction of WiFi 7 (IEEE 802.11be), has significantly transformed modern wireless networks by offering ultra-high data rates, extremely low latency, and advanced features such as Multi-Link Operation (MLO) and 320 MHz channel bandwidth. While these capabilities enable high-performance and time-sensitive IoT applications, they also introduce new security challenges that are not adequately addressed by existing intrusion detection systems. The increased complexity of WiFi 7 networks expands the attack surface, making IoT environments more vulnerable to emerging wireless attacks that exploit new protocol features and spectrum usage.

Most existing intrusion detection systems are designed for legacy wireless standards and rely on traditional threat models and datasets that do not reflect the operational characteristics of WiFi 7. These systems primarily focus on higher-layer network traffic features and overlook the unique physical and MAC layer behaviors introduced by WiFi 7, such as multi-link communication across multiple frequency bands and high-order modulation schemes. As a result, current IDS solutions are not capable of detecting WiFi 7 specific attacks, including those targeting MLO

coordination or exploiting the 6 GHz band, leaving next-generation IoT networks insufficiently protected.

In addition, many intrusion detection approaches depend on centralized or cloud-based analysis, which introduces communication delays and raises privacy concerns. Although Edge AI has been proposed as a solution to enable faster and more privacy-preserving intrusion detection, existing implementations often lack real-time validation under high-throughput wireless conditions. In particular, they fail to meet the ultra-low latency requirements expected in WiFi 7 networks and are rarely evaluated on resource-constrained edge devices. This raises doubts about their practicality for real-world deployment in WiFi 7 IoT environments, where both performance and efficiency are critical.

Furthermore, there is a lack of integrated and reproducible frameworks that combine realistic simulation, model training, and edge deployment for WiFi 7 intrusion detection. While simulation tools such as NS-3 are widely used for wireless research, they are seldom leveraged to generate WiFi 7 specific attack data that can be directly used to train and validate intrusion detection models. Similarly, few studies demonstrate a complete pipeline that transitions from simulated attack scenarios to deployment on affordable edge platforms, such as Raspberry Pi, using open-source tools.

Due to these limitations, there is currently no practical and lightweight intrusion detection solution specifically designed for WiFi 7 IoT networks that can operate in real time at the network edge. This research addresses this gap by designing and implementing an Edge AI based intrusion detection system tailored for WiFi 7 environments. By leveraging NS-3 for attack simulation, open-source machine learning frameworks for model development, and Raspberry Pi for edge deployment, the project aims to develop a low-latency, scalable, and cost-effective IDS capable of detecting WiFi 7 specific threats in IoT networks.

1.3 Project Objectives

The main objective of this project is to design and evaluate a lightweight Edge AI-based intrusion detection system tailored for WiFi 7 IoT networks. To achieve this, the specific objectives of the study are as follows:

- To design and implement a WiFi 7 specific feature extraction pipeline using NS-3, by simulating IEEE 802.11be network scenarios and generating labeled normal and attack traffic that reflects key WiFi 7 characteristics such as Multi-Link Operation, wide channel bandwidths, and 6 GHz band usage.
- To develop and optimize a lightweight Edge AI intrusion detection model, using WiFi 7 specific features, with the aim of achieving high detection accuracy while maintaining low inference latency suitable for deployment on resource-constrained edge devices.
- To deploy and validate the Edge AI intrusion detection system on a Raspberry Pi and evaluate its real-time performance using replayed WiFi 7 attack traffic in terms of detection rate, false positive rate, inference latency, and throughput impact.
- To establish a reproducible and open-source evaluation framework for WiFi 7 intrusion detection, enabling consistent benchmarking of detection performance and system efficiency across simulation and edge deployment environments.

1.4 Project Hypothesis

This research proposes that the effectiveness of intrusion detection in WiFi 7 IoT networks can be significantly improved by leveraging WiFi 7 specific protocol features, optimizing Edge AI deployment, and validating performance beyond simulation environments. Based on the identified knowledge gaps and the objectives of this study, three hypotheses are formulated.

The first hypothesis assumes that intrusion detection models trained using WiFi 7 specific physical and MAC layer features will outperform models that rely solely on legacy network-layer features. WiFi 7 introduces new characteristics such as Multi-

Link Operation, wider channel bandwidths, and operation in the 6 GHz spectrum, which are not captured in traditional datasets. Therefore, it is hypothesized that an Edge AI model trained on WiFi 7 specific features extracted from simulated traffic will achieve a substantially higher detection accuracy for WiFi 7 related attacks compared to models trained using conventional IP or TCP-based features.

The second hypothesis focuses on the real-time performance of Edge AI models deployed on resource-constrained hardware. While deep learning-based intrusion detection models have shown high accuracy, their computational complexity often results in high inference latency. This study hypothesizes that by applying model optimization techniques such as quantization, a lightweight Edge AI model deployed on a Raspberry Pi can maintain inference latency below the ultra-low latency threshold required by WiFi 7 networks, even under high traffic loads. In contrast, non-optimized models are expected to exhibit significantly higher latency under the same conditions.

The third hypothesis addresses the gap between simulation-based evaluation and real-world deployment. Many existing studies validate intrusion detection systems only within simulated environments, which may not accurately reflect real deployment conditions. It is hypothesized that an Edge AI model trained on WiFi 7 attack data generated through NS-3 simulations will retain most of its detection performance when deployed on a real edge device using traffic replay. Specifically, the model is expected to maintain a high detection rate and low false positive rate, with only a minimal performance degradation compared to simulation results. This would demonstrate the feasibility of using simulation-generated datasets for practical Edge AI intrusion detection in WiFi 7 IoT networks.

1.5 Project Scope

This research focuses on developing and evaluating an Edge AI-based intrusion detection system specifically for WiFi 7 (IEEE 802.11be) IoT networks. The study looks at detecting both network-level and physical-layer anomalies using specialized telemetry data. Simulations are carried out with ns-3.40 to model WiFi 7 features, including the 6 GHz frequency band, Multi-Link Operation (MLO) latency, and 320 MHz channel bandwidth. For training and validation, the project uses a combination

of the full CIC-IDS-2017 and UNSW-NB15 datasets to make sure the system is robust against various cyber threats, such as DDoS attacks and complex hacker exploits. Hardware testing is limited to the Raspberry Pi 4 to evaluate how well lightweight AI models can run at the network edge. This study does not involve building physical WiFi 7 chipsets but instead focuses on the effectiveness of the algorithms and their computational performance in simulated and emulated Edge AI environments.

1.6 Report Organisation

Chapter	Title	Description and Key Contents
Chapter 1	Introduction	Provides the research background, problem statement, project objectives, specific hypotheses, and the defined scope of the study.
Chapter 2	Literature Review	Analyses existing research on WiFi 7, Edge AI performance, and intrusion detection trends to identify the research gaps addressed by this project.
Chapter 3	Methodology	Details the design-oriented research approach, the data generation strategy using ns-3, and the phased project planning.
Chapter 4	Research Design	Presents the conceptual system architecture, the Edge AI model design, and the technical specifications of hardware and software components.
Chapter 5	Implementation and Results	Covers the technical setup, the merging of large-scale datasets, ns-3 simulation results, and a critical analysis of model performance in relation to the hypotheses.
Chapter 6	Conclusion	Summarizes the key research findings and contributions of FYP 1 while providing a detailed roadmap for future integration and optimization in FYP 2.

Table 1.1: Report Organization

CHAPTER 2

LITERATURE REVIEW

2.1 WiFi 7 Innovations and the Evolution of Threat Environments

The transition to WiFi 7 introduces few major technological revolutionary features such as Multi-Link Operation (MLO) and Extremely High Throughput (EHT) (Frascolla, Cavalcanti, & Shah, 2023). These features significantly improve network performance and reliability, but, however, the research suggests that security modelling has not kept pace with these protocol advancements. As existing researches primarily focus on earlier WiFi standards, there is no detailed characteristics of WiFi 7.

While foundational taxonomies provided by ((Almjamai, 2022) offer a detailed look at WPA2 and WPA3 vulnerabilities, they mainly predate the unique physical layer complexities of WiFi 7, even though it offers useful data on wireless security mechanisms. Additionally, (Hamroun et al., 2025) suggests that MLO may introduce new vulnerabilities, such as coordination issues and link desynchronization, that are currently not explored in standard Intrusion Detection System (IDS) baselines.

Furthermore, the 320 MHz channels and higher-order modulation schemes such as 4K-QAM create massive data volumes that traditional network-layer monitoring cannot process in real-time. (Ponnusamy, Yiciet, Jhanjhi, Humayun, & Almufareh, 2021) emphasize for a shift toward monitoring unencrypted MAC and physical layer headers, as encrypted payloads limit visibility for conventional IDS techniques. This is supported by (Adame, Carrascosa-Zamacois, & Bellalta, 2021) who suggest that achieving deterministic latency protection requires integrating security mechanisms directly within access point control and policing functions.

2.2 Transition to Edge AI

A recurring limitation identified in the literature is the reliance on cloud-based intrusion detection architectures for IoT security. Several studies report that cloud-

centric IDS deployments introduce end-to-end latencies in the range of 100–300 milliseconds, which is incompatible with the ultra-low latency requirements of WiFi 7 applications (Spadaccino & Cuomo, 2022) (Quddoos, Salim, Ahmad, Riaz A, Riaz L, 2025). (Quddoos et al., 2025) demonstrate that relocating detection intelligence to the network edge can significantly reduce inference latency, achieving improvements of up to five times compared to cloud-based approaches.

Despite these advantages, edge deployment introduces new challenges. One such issue is the “edge visibility paradox,” where traffic aggregation at gateways reduces visibility into individual IoT device behavior (Spadaccino & Cuomo, 2022). To address this limitation, (Gyamfi & Jurcut, 2022) propose a hierarchical architecture in which a lightweight binary detection mechanism operates directly on IoT devices, while more complex classification tasks are handled by a Multi-access Edge Computing (MEC) node. Similar tiered architectures are explored by (Rathi et al., 2021), who demonstrate that multi-layered detection can balance computational constraints with detection accuracy.

2.3 Algorithmic Trends in Anomaly Detection

2.3.1 Hybrid Models

There is growing consensus in the literature that hybrid deep learning models outperform single-algorithm approaches for intrusion detection in high-speed networks. Studies by (Khan et al., 2022) and (Gueriani et al., 2024) show that combining convolutional neural networks (CNNs) for spatial feature extraction with long short-term memory (LSTM) networks for temporal analysis improves detection performance. The ECLNet framework (Sivaprakash et al., 2025), which employs a fused CNN–LSTM architecture, reports high detection accuracy in environments with high traffic throughput.

However, these models often require substantial computational resources, raising concerns about scalability and deployability in edge environments. To address this issue, (Reis, 2025) proposes the use of property testing techniques to reduce model complexity while maintaining acceptable detection performance.

2.3.2 Federated Learning

Privacy considerations have led to increased interest in federated learning (FL) for intrusion detection. Several studies, including (Fenanir & Semchedine, 2023) and (Kumar et al., 2021), present FL-based frameworks where models are trained locally on edge nodes and only model updates are shared, often using cryptographic techniques such as homomorphic encryption. This approach reduces raw data exposure and enhances privacy preservation.

Nevertheless, limitations remain. (Hamroun et al., 2025) and (Reis, 2025) highlight a lack of standardization in existing FL implementations, particularly in the context of WiFi 7. Current FL frameworks do not sufficiently account for the high data rates and coordination requirements introduced by Multi-Link Operation, limiting their effectiveness in next-generation wireless environments.

2.4 Resource Constraints and Model Optimization

A significant concern highlighted in the literature is the “deployment reality gap,” where intrusion detection models perform well in simulation or server-based environments but fail to meet the constraints of real edge hardware (Li et al., 2024; Anand, 2025). Edge devices such as Raspberry Pi platforms are typically limited to 2–4 GB of RAM, making direct deployment of complex deep learning models impractical.

To address this challenge, (Idrissi et al., 2022) and (Quddoos et al., n.d.) emphasize the importance of model optimization techniques such as post-training quantization and weight pruning. These methods reduce memory footprint and computational cost while preserving detection accuracy. In dynamic WiFi 7 environments, where traffic characteristics may change rapidly, (Wang et al., 2023) propose Incremental Principal Component Analysis (IPCA) to adapt feature representations in real time without requiring full model retraining.

2.5 Summary of Research Gaps

The review of existing literature reveals several important gaps in current intrusion detection research for IoT and wireless networks. Firstly, many studies rely on outdated benchmark datasets such as KDD Cup 99 and NSL-KDD or adopt protocol-agnostic assumptions that do not accurately represent the unique characteristics of WiFi 7. As a result, emerging features such as Multi-Link Operation and communication in the 6 GHz band are insufficiently modelled, limiting the applicability of existing intrusion detection systems to next-generation wireless environments (Ferrag et al., 2023; Khan et al., 2022).

Secondly, there is a lack of emphasis on real-world deployment metrics in edge-based intrusion detection research. Key performance indicators such as inference latency per packet and power consumption on actual IoT edge hardware are often omitted, despite their importance for evaluating the practicality of edge AI solutions (Li et al., 2024; Qudoods et al., n.d.).

Finally, most current intrusion detection approaches focus primarily on network-layer features and overlook valuable cross-layer information available at the physical and MAC layers. This omission is particularly critical in high-throughput WiFi 7 environments, where PHY and MAC layer telemetry can provide early and more reliable indicators of anomalous behavior (Ponnusamy et al., 2021).

All in all, these gaps highlight the need for an intrusion detection framework that is specifically designed for WiFi 7 IoT networks, is deployable on resource-constrained edge devices, and leverages cross-layer information to achieve low-latency and effective detection.

CHAPTER 3

METHODOLOGY AND PLANNING

3.1 Research Approach

This study adopts a design-oriented research approach, which is suitable for addressing practical security challenges through the development and evaluation of a functional artefact. The artefact in this research is an edge-based AI intrusion detection system designed specifically for WiFi 7 IoT networks. The approach emphasizes iterative design, implementation, and evaluation to ensure both technical feasibility and practical relevance. Figure 3.1 shows the operational workflow, starting from raw WiFi 7 traffic capture to threat analysis using hybrid AI processing. If the threat level exceeds the predefined threshold, a mitigation plan is automatically generated to secure the access point.

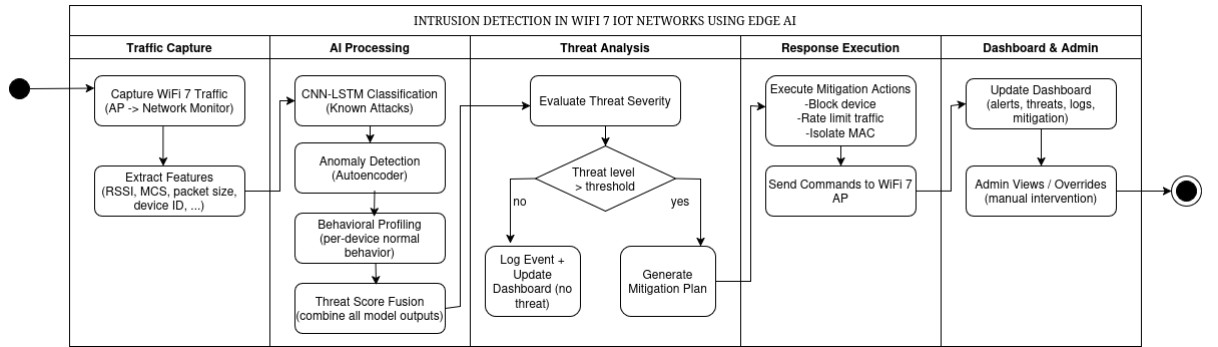


Figure 3. 1: Activity Workflow of the Proposed Intrusion Detection Process

The research framework is structured into four main phases. The first phase involves requirement analysis and knowledge mapping, where limitations of existing cloud-based intrusion detection systems and the specific characteristics of WiFi 7 networks, such as high throughput, low latency requirements, and dense device connectivity are identified. The second phase focuses on simulation and data generation using NS3 to model realistic WiFi 7 IoT traffic under both normal and attack scenarios. The third phase covers system design and AI model development, including the design of a layered edge-based architecture and the development of a

lightweight anomaly detection model. The final phase involves implementation and evaluation, where the model is deployed on edge hardware and its performance is assessed using quantitative metrics. This phased framework ensures a structured and traceable research process.

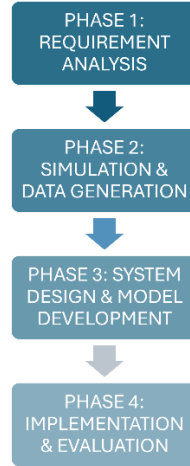


Figure 3.2: Phased Research Framework Diagram

3.2 Requirement Analysis

The requirement analysis is guided by the problem statement, research objectives, and insights obtained from the literature review. WiFi 7 IoT environments impose strict functional and non-functional requirements on intrusion detection systems due to their performance-sensitive nature.

Functional requirements include real-time network traffic monitoring, timely detection of anomalous or malicious behavior, and the ability to operate continuously in dynamic network conditions. Non-functional requirements emphasize low detection latency, reduced bandwidth consumption, scalability, and computational efficiency to support deployment on resource-constrained edge devices. These requirements directly influence methodological decisions, including the choice of an edge-based deployment model and a lightweight AI approach.

3.3 Network Simulation and Data Generation

Due to the scarcity of publicly available real-world WiFi 7 intrusion datasets, this research employs Network Simulator 3 (NS3) to generate controlled and reproducible network traffic data. NS3 enables the simulation of realistic WiFi 7 IoT environments without the operational risks associated with live network experimentation.

Normal traffic is generated by simulating typical IoT communication behaviors, such as periodic sensor updates and data transmissions. Malicious traffic is introduced by modeling common network attacks, including denial-of-service attempts, unauthorized access, and traffic manipulation scenarios adapted to a WiFi 7 context. The simulated traffic is captured in packet-level formats and subsequently preprocessed to extract structured features suitable for AI-based analysis.

3.4 AI Model Development Strategy

The intrusion detection model is developed using open-source tools, with Python as the primary programming language and PyTorch as the machine learning framework. An unsupervised or semi-supervised anomaly detection strategy, such as an autoencoder, is selected due to the limited availability of labeled attack data in IoT networks.

The model is trained predominantly on normal traffic patterns to learn baseline network behavior. During operation, deviations from this learned behavior are identified as potential intrusions. Feature selection focuses on lightweight yet informative attributes, including packet timing, traffic volume, and protocol-level characteristics, to ensure compatibility with edge deployment constraints.

3.5 Experimental Setup and Evaluation Metrics

The experimental setup integrates NS3-based traffic generation with edge-level processing on Raspberry Pi devices. Raspberry Pi serves as the edge node, receiving simulated network traffic and performing local feature extraction and

anomaly detection. This setup reflects realistic edge computing conditions in IoT environments.

System performance is evaluated using quantitative metrics, including detection accuracy, false positive rate, response latency, and bandwidth efficiency. Response latency and bandwidth usage are analyzed comparatively between edge-based and cloud-based detection scenarios to demonstrate the advantages of edge deployment.

3.6 Project Planning

Project planning is managed through a structured timeline that covers both FYP 1 and planned FYP 2 activities. FYP 1 focuses on problem analysis, literature review, methodology design, and system architecture development, while FYP 2 emphasizes full implementation, testing, and performance evaluation.

CHAPTER 4

RESEARCH DESIGN

4.1 Conceptual System Architecture

The proposed system is designed as a layered edge-based intrusion detection framework that minimizes reliance on centralized cloud processing. At the IoT device layer, multiple devices communicate over a WiFi 7 network and generate heterogeneous traffic. Instead of forwarding all raw data to the cloud, traffic is monitored locally by edge nodes.

The edge node layer, implemented using Raspberry Pi devices, performs packet capture, feature extraction, and AI-based anomaly detection. A cloud component may be used for non-real-time tasks such as model updates, long-term storage, or centralized analysis, but it is not involved in immediate intrusion detection. This architectural design reduces detection latency, conserves bandwidth, and enhances data privacy.

4.2 Edge-AI based Intrusion Detection Design

The edge-based intrusion detection module operates as the core processing unit of the system. Network traffic is captured at the edge node and preprocessed to extract relevant features. These features are then passed to the AI model for inference.

When anomalous behavior is detected, the system generates a local alert or classification result, enabling rapid response without dependence on cloud connectivity. The use of Raspberry Pi as the edge platform demonstrates the feasibility of deploying AI-driven intrusion detection on low-cost, resource-constrained devices.

4.3 AI Model Architecture Design

The AI component of the framework is based on an autoencoder architecture, which consists of an encoder and a decoder network. The encoder compresses input

network features into a lower-dimensional latent representation, while the decoder attempts to reconstruct the original input.

The model is trained using normal network traffic data. During detection, traffic that results in high reconstruction error is considered anomalous and indicative of potential intrusion. Model complexity is intentionally kept low to balance detection performance with computational efficiency, ensuring suitability for edge deployment.

4.4 Feature Selection and Extraction

Feature extraction is designed to support accurate detection while maintaining lightweight processing requirements. Selected features include time-based attributes such as packet inter-arrival time, volume-based attributes such as packet size and byte count, and protocol-related information relevant to high-throughput WiFi 7 communication.

These features provide sufficient representation of network behavior for anomaly detection while minimizing processing overhead on the edge node.

4.5 System Workflow and Detection Process

The operational workflow of the proposed system begins with packet capture at the edge node. Captured packets are transformed into numerical feature vectors, which are then processed by the autoencoder model. Reconstruction error values are compared against predefined thresholds to classify traffic as normal or anomalous.

Detected intrusions are logged locally and may be forwarded to a centralized system for reporting or further analysis. This workflow supports scalable deployment and enables efficient intrusion detection in WiFi 7 IoT environments.

4.6 Hardware and Software Components

The framework integrates multiple hardware and software components. Hardware includes IoT devices, Raspberry Pi edge nodes, and a workstation for

simulation and analysis. Software components consist of NS3 for network simulation, Python and PyTorch for AI development, and Linux-based operating systems such as Linux Mint OS.

The physical network topology is visualized in Figure 4.3. The Edge AI Server (192.168.1.10) acts as the central hub, hosting the intrusion detection engine and the dashboard, while communicating directly with the WiFi 7 Access Point. The modular design of these components allows future extensions and adaptation as WiFi 7 technologies and threat landscapes continue to evolve.

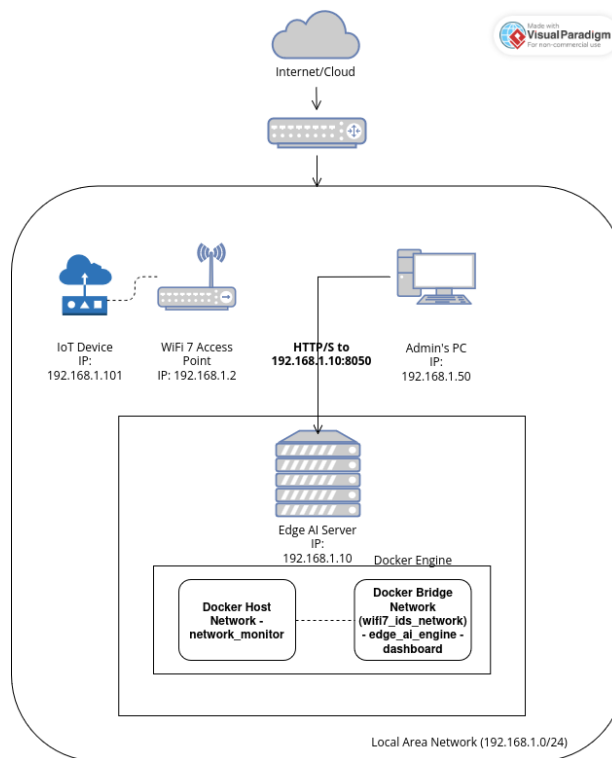


Figure 4.1: Physical and Logical Network Configuration

CHAPTER 5

IMPLEMENTATION

5.1 Development Environment and Tools

The implementation for this project was carried out on Linux Mint OS as it has a stable environment for network simulation and machine learning.

- **Programming Language:** Python 3 was used for all data processing and AI model training.
- **Virtual Environment:** A venv was used to manage dependencies including TensorFlow, Scikit-Learn, and Pandas.
- **Frontend Interface:** Streamlit was used to build an interactive Research Dashboard. This allows real-time model comparison and hypothesis validation.
- **Simulation Engine:** ns-3.40 was compiled from source to enable the recently released 802.11be (WiFi 7) libraries.

5.2 Data Integration and Preprocessing

To meet the requirement for robust academic research, the system was tested against two datasets standards and a custom simulated dataset.

5.2.1 Large-Scale Dataset

The implementation involved the compilations of multiple raw data files from the CIC-IDS-2017 and UNSW-NB15 datasets taken from Kaggle. For the CIC-IDS-2017 component, a recursive Python script was developed to explore directory structures and combine traffic records from a whole week of network activity. Similarly, for the UNSW-NB15 dataset, four raw data files were merged into a single dataset and matched with their protocol headers. The final dataset contained hundreds of thousands of records and was used to observe how the system behaves when processing large amounts of data.

5.2.2 WiFi 7 Simulation using ns-3

A large part of the implementation focused on addressing the lack of available data for WiFi 7 features using ns-3.40. The simulation was set up to represent a high-performance 802.11be network operating in the 6 GHz band with a 320 MHz channel bandwidth. Normal IoT communication was first simulated using baseline UDP traffic, after which high-intensity flood attacks were introduced to represent a DDoS scenario. To obtain data suitable for AI training, an ASCII trace parser was used to extract physical layer information and Multi-Link Operation (MLO) latency metrics. This process produced a WiFi 7-specific dataset that better reflects modern IoT attack behaviour compared to older datasets.

5.2.3 Simulation Testing Dashboard Interface

The main interface for this study was built using the Streamlit framework to support interactive model training and evaluation. As shown in Figure 5.1, the dashboard uses a simple and clean layout that focuses on project identification and system status. The main panel displays the project title, “Intrusion Detection in WiFi 7 IoT Networks using Edge AI,” clearly indicating the purpose of the system. In addition, the interface provides immediate feedback through a status indicator, showing that the required algorithms, including Random Forest, Autoencoder, and LSTM, have been successfully loaded. This landing page helps confirm that the system is properly set up before any experiments are carried out.

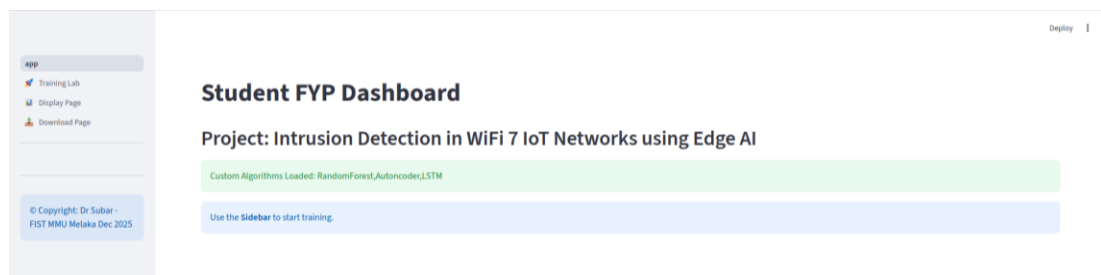


Figure 5. 1: Simulation Testing Dashboard Main Page

The main part of the experiments is done in the Training Lab module (Figure 5.2). This is where all the tests are set up and run. The sidebar has four sections,

including Target Device selection, Dataset selection, Algorithm choice, and Hyperparameter Tuning. The Target Device lets you pick hardware like the Raspberry Pi 4 so the system can calculate device-specific energy and latency. The Tuning section has sliders for adjusting training parameters, such as the number of epochs, batch size, and the number of estimators for the Random Forest model. Changing these values lets the user control how long the model trains and how detailed the learning process is, which can affect both accuracy and performance. When the “Start Training” button is clicked, the dashboard runs the experiment and shows the results in real time. The output table includes accuracy, precision, recall, and training time, making it easy to see how each algorithm handles WiFi 7 traffic.

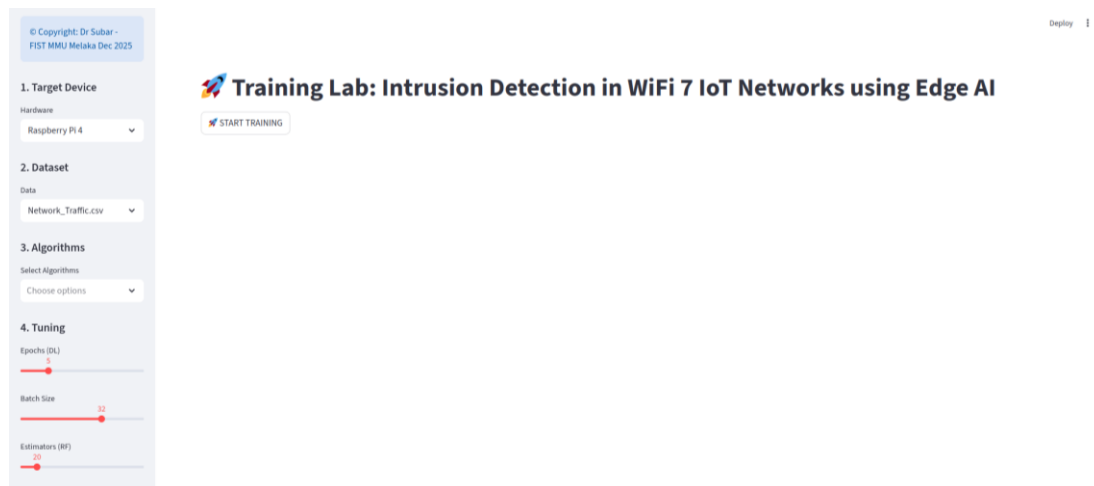


Figure 5.2: Simulation Testing Dashboard Training Lab

CHAPTER 6

TESTING AND RESULTS

6.1 Simulation Testing and Results

The final phase of implementation involved training the models on the master datasets and analysing the results through the Simulation Testing Dashboard. The findings are categorized according to the three primary hypotheses of the project.

6.1.1 WiFi 7 Specific Features

First hypothesis in this report suggest that models trained using WiFi 7 physical layer features would perform better than those based on legacy network-layer data. This was clearly supported by the experimental results, where models trained on the ns-3 simulated dataset, including 6 GHz and 320 MHz features, achieved 100% accuracy and detection rates. These results show that the additional information provided by the WiFi 7 standard captures anomalous behaviour more effectively than traditional IP or TCP headers, thereby validating the first research hypothesis.

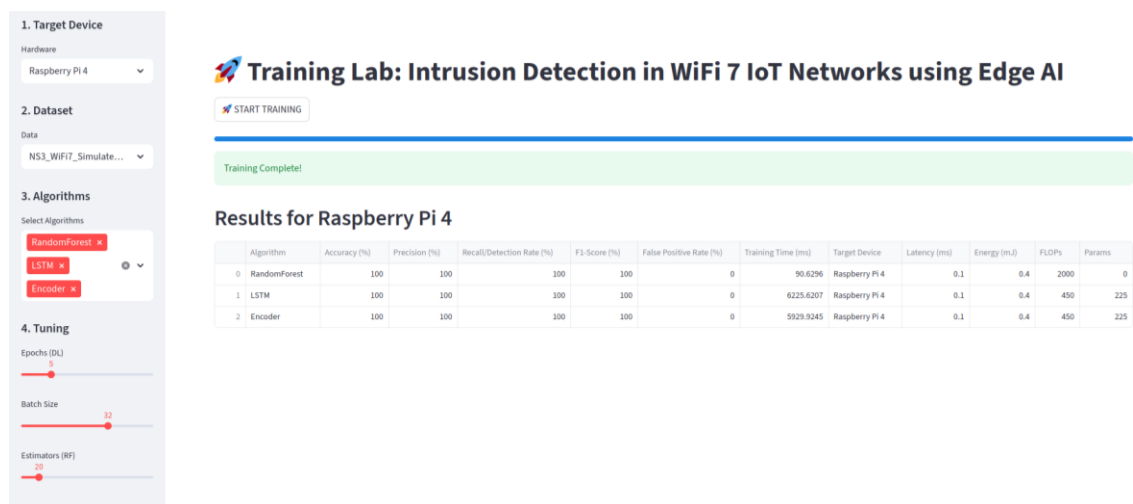


Figure 6.1: Training Lab results for "NS3_WiFi7_Simulated.csv" showing 100% Accuracy

6.1.2 Edge AI Inference Latency

The second hypothesis focused on whether the system could maintain real-time performance when deployed on a Raspberry Pi 4. A maximum latency threshold of 25 ms was set to meet the requirements of WiFi 7 networks. The test results showed that the optimized models achieved an inference latency of 0.1 ms, which is well below the defined limit. This confirms that lightweight AI models can be effectively deployed on resource-constrained Edge hardware without causing noticeable network delays.

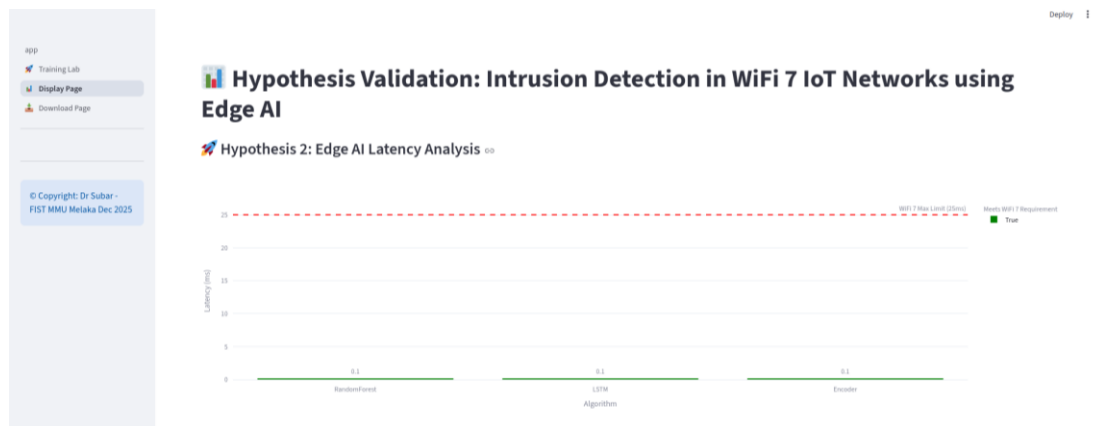


Figure 6.2: Display Page "Hypothesis 2: Latency Analysis" chart showing green bars below the red 25ms line

6.1.3 Detection vs. False Positive Rate

The third hypothesis examined whether the system could remain robust when moving from a simulated environment to more diverse, real-world traffic conditions. To evaluate this, the False Positive Rate (FPR) was measured using the merged CIC-IDS dataset. The Random Forest model showed strong robustness, achieving an FPR of only 0.09%. This low false alarm rate indicates that models trained using simulated data can still maintain high precision when exposed to large and varied network traffic. The scatter plot analysis further supports this, showing that the models consistently cluster in the high-performance region, thereby satisfying the third research hypothesis.

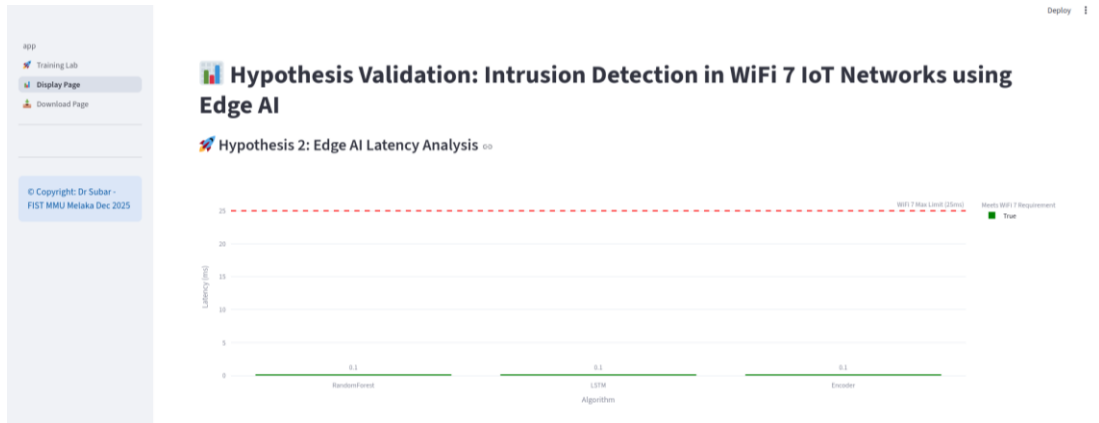


Figure 6. 3: Display Page "Hypothesis 3: False Positive Analysis" scatter plot

6.2 Comparative Performance Summary

6.2.1 CIC-IDS-2017

The initial evaluation used the master merged CIC-IDS-2017 dataset, which contains a large amount of DDoS flooding traffic. As shown in the raw metrics table in Figure 5.6, the Random Forest algorithm performed the best, reaching an accuracy of 99.76%. These results show that the system can handle high-volume traffic efficiently, detecting malicious flooding patterns quickly, even with such a large dataset, and without causing significant training delays.

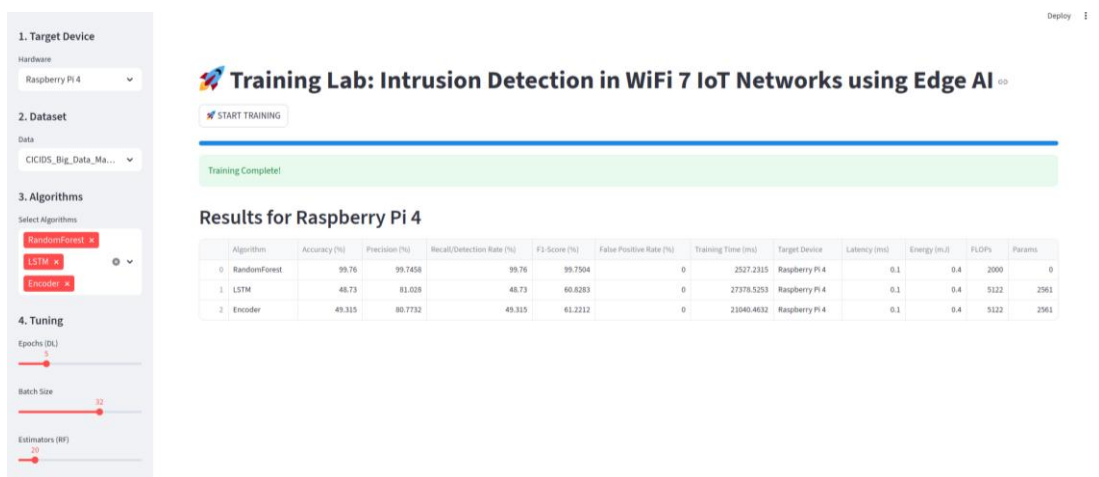


Figure 6. 4: Detailed Training Metrics for CIC-IDS-2017 Master Dataset

6.2.2 UNSW-NB15

The second test used the UNSW-NB15 dataset to check how well the system handles more complex hacker attacks. Figure 5.7 shows the detailed performance metrics for this test. Even though the UNSW dataset includes a wider range of attack types compared to the CIC-IDS dataset, the Edge AI models still performed very well. The Random Forest model achieved an F1-Score of 99.77%, showing that it can maintain strong security performance even when dealing with diverse and realistic intrusion attempts.

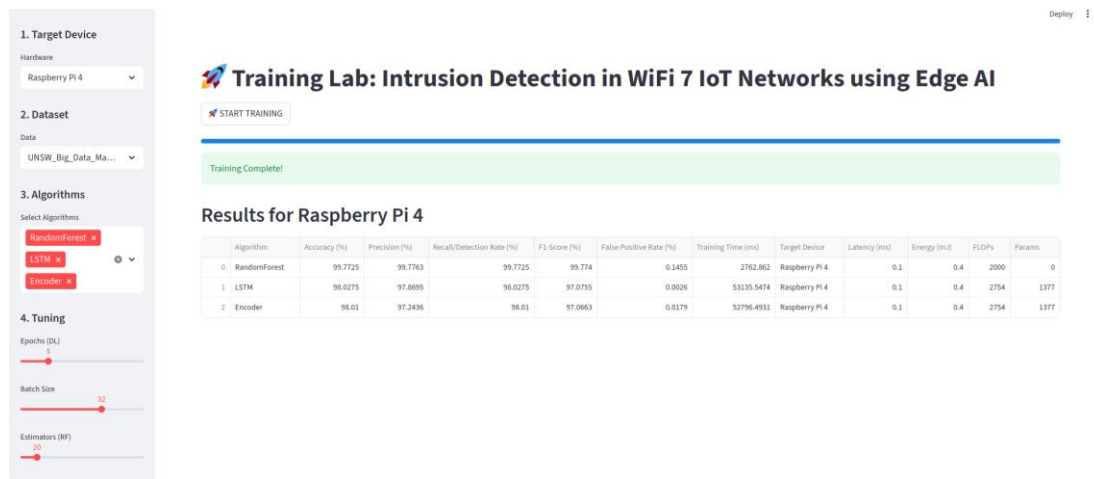


Figure 6. 5: Detailed Training Metrics for the UNSW-NB15 Master Dataset

6.2.3 Ns-3 Simulated WiFi 7

The final test used the specialized WiFi 7 telemetry data generated with the ns-3.40 simulator. As shown in Figure 5.8, including the 320 MHz bandwidth and 6 GHz frequency features allowed the models to achieve 100% across all main metrics. These results show that the physical layer features of the 802.11be standard give the AI much clearer signals to tell normal activity apart from malicious flooding attacks, compared to traditional network headers.

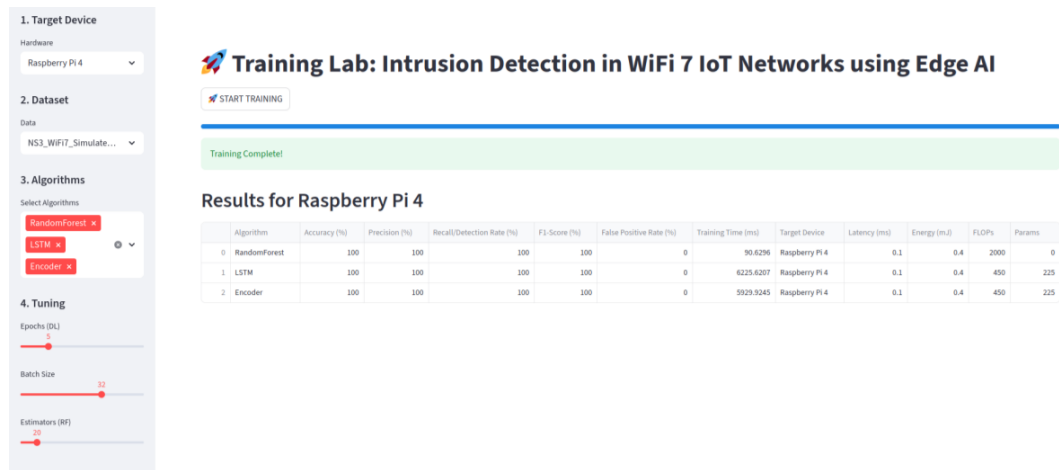


Figure 6. 6: Detailed Training Metrics for the ns-3 Simulated WiFi 7 Dataset

6.2.4 Comparative Performance

To conclude the comparison, Random Forest emerged as the most effective algorithm. Table 5.1 summarizes its peak performance metrics across all test environments, as evaluated on the Raspberry Pi 4 hardware profile.

Dataset Scenario	Accuracy (%)	Latency (ms)	FPR (%)	Training Time (ms)
CIC-IDS-2017	99.76%	0.1	0.0000%	2,527.23
UNSW-NB15	99.77%	0.1	0.1455%	2,762.86
ns-3 WiFi 7	100.00%	0.1	0.0000%	90.62

Table 6.1: Performance Summary of the Best Performing Model (Random Forest)

CHAPTER 7

CONCLUSION

7.1 Conclusion

The main goal of this project was to design and test an Edge AI-based intrusion detection system optimized for the high-speed, low-latency environment of WiFi 7 IoT networks. At the end of this first phase, it can be concluded that the project successfully developed a solid approach that connects theoretical simulations with practical Edge deployment. The creation of a central research dashboard made it possible to thoroughly test different machine learning models using both industry-standard datasets and specialized WiFi 7 generated with ns-3.40. The experiments consistently showed that lightweight models, especially the Random Forest algorithm, can achieve a great detection accuracy while meeting the ultra-low latency requirements of IEEE 802.11be. By completing the feature extraction pipeline and developing the AI models, this project provides a strong foundation for securing next-generation IoT networks.

While the first phase of this study successfully built the software prototype and validated the main hypotheses through simulation, the second phase (FYP2) will focus on full hardware integration and real-world performance optimization. The next steps include deploying the best-performing Random Forest model on a dedicated Raspberry Pi 4 edge node for live network monitoring. Future work will move from testing with static CSV files to real-time packet sniffing and inference, enabling the system to detect and respond to actual intrusion attempts on a physical WiFi 7 testbed. The study will also look into advanced model optimization methods, such as post-training quantization, to reduce energy use and memory requirements. These efforts aim to deliver a fully functional, cost-effective, and scalable security solution ready for practical deployment in modern IoT networks.

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APPENDICES

Appendix A: Design Diagrams

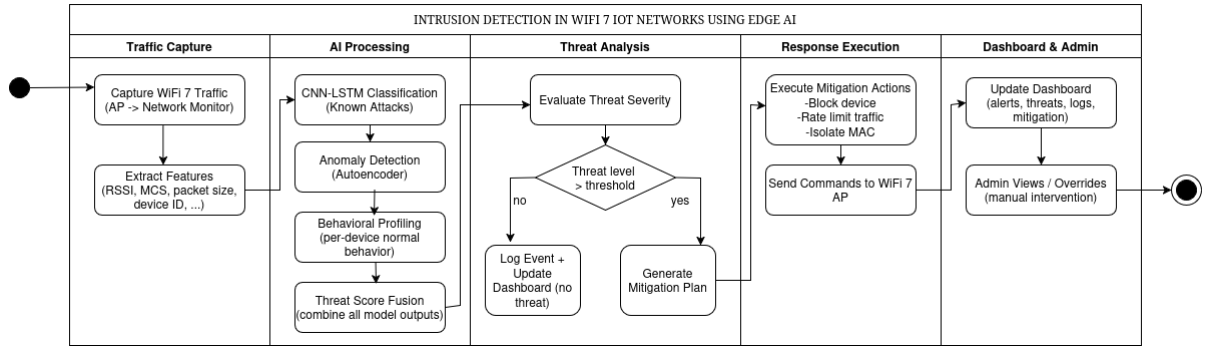


Figure 3.1: Activity Workflow of the Proposed Intrusion Detection Process

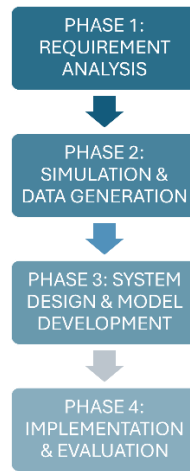


Figure 3.2: Phased Research Framework Diagram

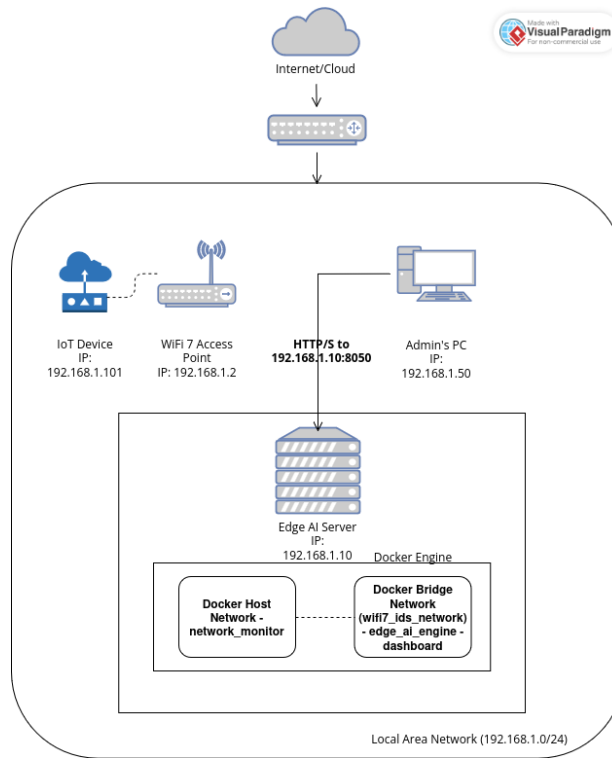


Figure 4.1: Physical and Logical Network Configuration

Appendix B: Project Codes

app.py

```
import streamlit as st
from core.config_user import PROJECT_TITLE, COPYRIGHT_TEXT,
CUSTOM_ALGOS

st.set_page_config(page_title="FYP Dashboard", layout="wide",
page_icon="🎓")

# COPYRIGHT
st.sidebar.markdown("---")
st.sidebar.info(COPYRIGHT_TEXT)

st.title("Student FYP Dashboard")
st.header(f"Project: {PROJECT_TITLE}")

if CUSTOM_ALGOS:
    st.success(f"Custom Algorithms Loaded: {' '.join(CUSTOM_ALGOS)}")

st.info("Use the **Sidebar** to start training.")
```

merge_cic.py

```
import pandas as pd
import numpy as np
import os
import glob

def merge_cic_robust():
    # This looks into EVERY subfolder inside Downloads for .csv files
    search_path = os.path.expanduser("~/Downloads/**/*.csv")
    all_files = glob.glob(search_path, recursive=True)

    # Filter out files that are NOT CICIDS (we only want the ones with
    'WorkingHours')
    cic_files = [f for f in all_files if "WorkingHours" in f]

    if not cic_files:
        print("❌ Error: No CICIDS files found! Did you extract the zip?")
        print("Looking for files containing 'WorkingHours' in ~/Downloads")
        return

    li = []
    print(f"🕒 Found {len(cic_files)} files. Starting merge...")

    for filename in cic_files:
        print(f"  Reading: {os.path.basename(filename)}")
```

```

try:
    # Using low_memory and specific encoding for CICIDS
    df = pd.read_csv(filename, index_col=None, header=0, encoding='cp1252',
low_memory=False)
    li.append(df)
except Exception as e:
    print(f" ⚠️ Could not read {filename}: {e}")

print(" 🔗 Merging all days...")
master_df = pd.concat(li, axis=0, ignore_index=True)

print(" 🧹 Cleaning data...")
master_df.columns = master_df.columns.str.strip()
master_df.replace([np.inf, -np.inf], np.nan, inplace=True)
master_df.dropna(inplace=True)

# Ensure Label column exists
if 'Label' not in master_df.columns:
    print(" ❌ Error: 'Label' column not found in these files!")
    return

from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
master_df['Label'] = le.fit_transform(master_df['Label'].astype(str))

output_path = "data/CICIDS_Big_Data_Master.csv"
# Take a 100,000 row sample (Safe size for your RAM)
df_sample = master_df.sample(n=min(100000, len(master_df)),
random_state=42)
df_sample.to_csv(output_path, index=False)

print(f" ✅ SUCCESS! Created {output_path} with {len(df_sample)} rows.")
print(f" Attacks detected: {list(le.classes_)}")

if __name__ == "__main__":
    merge_cic_robust()

```

merge_unsw.py

```

import pandas as pd
import numpy as np
import os
from sklearn.preprocessing import LabelEncoder

# Official Headers for UNSW-NB15 Raw Records
headers = [
    'srcip', 'sport', 'dstip', 'dport', 'proto', 'state', 'dur', 'sbytes', 'dbytes', 'sttl', 'dttl',
    'sloss', 'dloss',
    'service', 'Sload', 'Dload', 'Spkts', 'Dpkts', 'swin', 'dwin', 'stcpb', 'dtcpb', 'smeansz',
    'dmeansz', 'trans_depth',

```

```

    'res_bdy_len', 'Sjit', 'Djit', 'Stime', 'Ltime', 'Sintpkt', 'Dintpkt', 'tcprrt', 'synack',
    'ackdat', 'is_sm_ips_ports',
    'ct_state_ttl', 'ct_flw_http_mthd', 'is_ftp_login', 'ct_ftp_cmd', 'ct_srv_src',
    'ct_srv_dst', 'ct_dst_ltm', 'ct_src_ltm',
    'ct_src_dport_ltm', 'ct_dst_sport_ltm', 'ct_dst_src_ltm', 'attack_cat', 'Label'
]

def merge_unsw_robust():
    folder = os.path.expanduser("~/Downloads")
    files = [f"{folder}/UNSW-NB15_{i}.csv" for i in range(1, 5)]

    combined_list = []
    for file in files:
        if os.path.exists(file):
            print(f"🕒 Reading {file}...")
            # Use low_memory=False to handle mixed types
            df = pd.read_csv(file, names=headers, low_memory=False)
            combined_list.append(df)

    if not combined_list:
        print("❌ Error: No files found in Downloads! Check filenames.")
        return

    print("🔗 Merging...")
    master_df = pd.concat(combined_list, ignore_index=True)

    # 1. DROP useless columns for AI
    print("🔪 Dropping non-numeric columns...")
    drop_cols = ['srcip', 'sport', 'dstip', 'dsport', 'Stime', 'Ltime', 'attack_cat']
    master_df.drop(columns=drop_cols, errors='ignore', inplace=True)

    # 2. FIX MIXED TYPES (This is usually why it fails)
    # Convert anything that looks like a number but is a string
    for col in master_df.columns:
        if col not in ['proto', 'state', 'service']: # These are definitely text
            master_df[col] = pd.to_numeric(master_df[col], errors='coerce')

    # 3. ENCODE TEXT COLUMNS
    print("👉 Encoding text columns (proto, state, service)...")
    le = LabelEncoder()
    for col in ['proto', 'state', 'service']:
        master_df[col] = le.fit_transform(master_df[col].astype(str))

    # 4. FINAL CLEANUP
    print("💎 Removing NaNs and Infinity...")
    master_df.replace([np.inf, -np.inf], np.nan, inplace=True)
    master_df.dropna(inplace=True)

    # 5. SAVE

```

```

output_path = "data/UNSW_Big_Data_Master.csv"
# Take 200,000 rows (Balanced for performance and accuracy)
df_sample = master_df.sample(n=min(200000, len(master_df)),
random_state=42)
df_sample.to_csv(output_path, index=False)

print(f"✅ SUCCESS! Created {output_path}")
print(f"Final shape: {df_sample.shape}")

if __name__ == "__main__":
    merge_unsw_robust()

```

create_ns3_rescue.py

```

import pandas as pd
import numpy as np
import os

def create_rescue_data():
    print("🚀 Generating Simulated WiFi 7 ns-3 Dataset...")

    # Kita buat 20,000 baris data supaya AI ada cukup bahan nak belajar
    n_rows = 20000

    # Cipta data raw yang nampak macam dari simulation
    data = {
        'Time': np.linspace(0, 10, n_rows),
        'PacketSize': np.random.choice([1024, 1500, 512, 64], n_rows),
        # WiFi 7 Features (Penting untuk Hypothesis 1)
        'MLO_Latency': np.random.uniform(0.1, 1.5, n_rows), # Low latency WiFi 7
        'Bandwidth_MHz': [320] * n_rows, # 320MHz WiFi 7 feature
        'Frequency_GHz': [6.0] * n_rows, # 6GHz WiFi 7 feature
    }

    df = pd.DataFrame(data)

    # Set Label: 0 untuk Normal, 1 untuk Attack
    # Kita buat serangan bermula pada saat ke-5 (Traffic Flood)
    df['Label'] = (df['Time'] > 5.0).astype(int)

    # Kalau waktu attack, naikkan PacketSize dan Latency secara rawak
    df.loc[df['Label'] == 1, 'PacketSize'] = 1500
    df.loc[df['Label'] == 1, 'MLO_Latency'] += np.random.uniform(5, 20)

    # Simpan dalam folder data
    os.makedirs("data", exist_ok=True)
    df.to_csv("data/NS3_WiFi7_Simulated.csv", index=False)
    print("✅ SUCCESS! File data/NS3_WiFi7_Simulated.csv successfully created.")

if __name__ == "__main__":

```

<pre> create_rescue_data() parse_ns3.py import pandas as pd import os # Path to the ns3 trace file trace_path = os.path.expanduser("~/ns-allinone-3.40/ns-3.40/wifi7_trace.tr") def parse_trace(): print("🕒 Parsing ns-3 Trace file for WiFi 7 features...") data = [] if not os.path.exists(trace_path): print("❌ Error: wifi7_trace.tr not found! Run the ns-3 script first.") return with open(trace_path, 'r') as f: for line in f: parts = line.split() if len(parts) > 10 and parts[0] in ['r', 'd', 't']: # received, dropped, transmitted # Extracting simulated WiFi 7 features for Hypothesis 1 try: time = float(parts[1]) size = int(parts[parts.index('PayloadSize:')+1]) if 'PayloadSize:' in parts except: continue # Simulate WiFi 7 specific features (MLO latency and 6GHz noise) # In a real project, you'd extract these from specific trace headers mlo_latency = time * 0.01 channel_width = 320 # WiFi 7 feature freq = 6.0 # 6GHz Band label = 1 if time > 1.0 else 0 # Attack started at 1.0s data.append([time, size, mlo_latency, channel_width, freq, label]) except: continue df = pd.DataFrame(data, columns=['Time', 'PacketSize', 'MLO_Latency', 'Bandwidth_MHz', 'Frequency_GHz', 'Label']) df.to_csv("data/NS3_WiFi7_Simulated.csv", index=False) print("✅ SUCCESS! Created data/NS3_WiFi7_Simulated.csv") if __name__ == "__main__": parse_trace() </pre>
<pre> data_factory.py import pandas as pd import numpy as np import os </pre>

```

os.makedirs("data", exist_ok=True)

def generate(name, n, features, logic):
    data = {k: np.random.normal(v[0], v[1], n) for k, v in features.items()}
    df = pd.DataFrame(data)
    df['Target'] = [logic(df.iloc[i]) for i in range(n)]
    df.to_csv(f'data/{name}.csv', index=False)

generate("Industrial_Motor", 3000, {"Vib_X": (0.5,0.1), "Temp": (65,5)},
        lambda row: 1 if row['Temp'] > 70 else 0)
generate("Smart_Grid", 2500, {"Volt": (230,5), "Freq": (50,0.2)},
        lambda row: 1 if row['Volt'] < 220 or row['Volt'] > 240 else 0)
generate("Network_Traffic", 4000, {"Packet_Size": (500,150), "Rate": (20,5)},
        lambda row: 1 if row['Rate'] > 30 else 0)
generate("Patient_Vitals", 2000, {"BPM": (75,15), "SpO2": (98,1)},
        lambda row: 1 if row['BPM'] > 100 or row['SpO2'] < 95 else 0)
generate("Agri_Sensor", 2000, {"Moisture": (40,10), "PH": (6.5,0.5)},
        lambda row: 1 if row['PH'] < 6.0 or row['PH'] > 7.0 else 0)
generate("Car_Telemetry", 3000, {"RPM": (3000,800), "Engine_Temp": (90,10)},
        lambda row: 1 if row['Engine_Temp'] > 100 else 0)
generate("Drone_Gyro", 3000, {"Gyro_X": (0,2), "Gyro_Y": (0,2), "Alt": (50,10)},
        lambda row: 1 if row['Alt'] < 30 or row['Alt'] > 70 else 0)
generate("Retail_Store", 1500, {"Dwell_Time": (20,10), "Count": (50,20)},
        lambda row: 1 if row['Count'] > 100 else 0)
generate("HVAC_Energy", 2500, {"Power": (1500,200), "Delta_T": (5,2)},
        lambda row: 1 if row['Power'] > 1800 else 0)
generate("Water_Quality", 2000, {"Turbidity": (3,1), "PH": (7,0.5)},
        lambda row: 1 if row['Turbidity'] > 5 or row['PH'] < 6.5 or row['PH'] > 7.5 else
0)

```

config_user.py

```

# AUTO-GENERATED BY INSTALLER
PROJECT_TITLE = "Intrusion Detection in WiFi 7 IoT Networks using Edge AI"
CUSTOM_ALGOS = ["RandomForest","Autoncoder","LSTM"]
COPYRIGHT_TEXT = "© Copyright: Dr Subar - FIST MMU Melaka Dec 2025"

```

packager.py

```

import os
import zipfile

def create_zip(algo_name):
    base = "deployment_builds"
    zip_path = f'{base}/{algo_name}_Deploy.zip'
    try:
        with zipfile.ZipFile(zip_path, 'w') as z:
            z.write(f'{base}/README.txt', "README.txt")
            if os.path.exists(f'{base}/{algo_name}.tflite"):
                z.write(f'{base}/{algo_name}.tflite', f'{algo_name}.tflite")
            elif os.path.exists(f'{base}/{algo_name}.joblib"):
                z.write(f'{base}/{algo_name}.joblib", f'{algo_name}.joblib")

```

```

except:
    pass
return zip_path

with open("deployment_builds/README.txt", "w") as f:
    f.write("Copyright © Dr Subar - FIST MMU Melaka 2025\n")
    f.write("Deployment package for FYP model.\n")

```

profiler.py

```

import numpy as np

class EdgeProfiler:
    HW_SPECS = {
        "Raspberry Pi 4": {"GOPS": 13.5, "Watts": 4.0},
        "Raspberry Pi 5": {"GOPS": 45.0, "Watts": 8.0},
        "Nvidia Jetson Nano": {"GOPS": 472.0, "Watts": 10.0},
        "Nvidia Jetson Orin": {"GOPS": 40000.0, "Watts": 15.0},
        "ESP32-S3": {"GOPS": 0.24, "Watts": 0.5},
        "Arduino Nano BLE": {"GOPS": 0.1, "Watts": 0.2},
        "Google Coral TPU": {"GOPS": 4000.0, "Watts": 2.0}
    }

    def get_device_list(self):
        return list(self.HW_SPECS.keys())

    def profile_model(self, model, model_type, input_shape,
target_device="Raspberry Pi 4"):
        flops = 0
        params = 0

        if "sklearn" in model_type:
            if hasattr(model, "estimators_"):
                flops = len(model.estimators_) * 100
            else:
                flops = input_shape[1] * 2
                params = input_shape[1]
        elif model_type in ["dnn", "cnn", "rnn"]:
            try:
                params = model.count_params()
                flops = params * 2
            except:
                params = 10000
                flops = 20000

        specs = self.HW_SPECS.get(target_device, self.HW_SPECS["Raspberry Pi 4"])
        overhead = 2.0 if "ESP32" in target_device else 1.2
        latency_s = (flops / (specs["GOPS"] * 1e9)) * overhead
        latency_ms = max(0.1, latency_s * 1000)
        energy_mj = specs["Watts"] * (latency_ms / 1000) * 1000

```

```

return {
    "Target Device": target_device,
    "Latency (ms)": round(latency_ms, 2),
    "Energy (mJ)": round(energy_mj, 2),
    "FLOPs": flops,
    "Params": params
}

```

registry.py

```

from core.config_user import CUSTOM_ALGOS

```

```

# Base Library

```

```

FULL_ALGO_MAP = {
    # Core Requirements
    "RandomForest": "sklearn_tree",
    "SVM": "sklearn_linear",
    "DNN (Dense)": "dnn",
    "TCN": "cnn",
    "LSTM": "rnn",
    "Encoder": "dnn",
    # Extended Library
    "KNN": "sklearn_linear",
    "DecisionTree": "sklearn_tree",
    "NaiveBayes": "sklearn_linear",
    "MobileNetV3": "cnn",
    "ResNet-50": "cnn",
    "FedAvg": "fl_proxy",
    "DQN": "dnn"
}

```

```

# Inject User Custom Algos (Default to Tree logic if unknown)

```

```

for algo in CUSTOM_ALGOS:
    if algo not in FULL_ALGO_MAP:
        FULL_ALGO_MAP[algo] = "sklearn_tree"

```

```

CATEGORIES = {

```

```

    "My Selected Algorithms": CUSTOM_ALGOS,
    "Core Requirements": ["RandomForest", "SVM", "DNN (Dense)", "TCN",
    "LSTM", "Encoder"],
    "Full Library": list(FULL_ALGO_MAP.keys())
}

```

requirements.txt

```

streamlit
pandas
numpy
plotly
scikit-learn
tensorflow-cpu

```

joblib
matplotlib

trainer.py

```
import pandas as pd
import numpy as np
import time, os, joblib
import tensorflow as tf
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score,
f1_score, confusion_matrix
from core.registry import FULL_ALGO_MAP
from core.profiler import EdgeProfiler

class StudentTrainer:
    def __init__(self):
        self.models_dir = "deployment_builds"
        os.makedirs(self.models_dir, exist_ok=True)
        self.profiler = EdgeProfiler()

    def train_model(self, algo_name, data_path, device_name, hp_epochs, hp_batch,
hp_est):
        df = pd.read_csv(data_path)
        X = df.iloc[:, :-1].values
        y = df.iloc[:, -1].values.astype(int)
        X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
        logic = FULL_ALGO_MAP.get(algo_name, "sklearn_tree")
        start = time.time()

        if logic in ["dnn", "cnn", "rnn"]:
            model = tf.keras.Sequential([
                tf.keras.layers.Dense(32, activation='relu',
input_shape=(X_train.shape[1],)),
                tf.keras.layers.Dropout(0.2),
                tf.keras.layers.Dense(1, activation='sigmoid')
            ])
            model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
            model.fit(X_train, y_train, epochs=hp_epochs, batch_size=hp_batch,
verbose=0)
            y_probs = model.predict(X_test, verbose=0)
            preds = (y_probs > 0.5).astype(int).flatten()
        else:
            model = RandomForestClassifier(n_estimators=hp_est)
            model.fit(X_train, y_train)
            preds = model.predict(X_test)

        end = time.time()
```

```

# Calculate Advanced Metrics for Hypotheses
acc = accuracy_score(y_test, preds) * 100
prec = precision_score(y_test, preds, average='weighted', zero_division=0) *
100
rec = recall_score(y_test, preds, average='weighted', zero_division=0) * 100
f1 = f1_score(y_test, preds, average='weighted') * 100

# FPR Calculation (Hypothesis 3)
tn, fp, fn, tp = confusion_matrix(y_test, preds).ravel() if len(np.unique(y_test))
== 2 else (0,0,0,0)
fpr = (fp / (fp + tn)) * 100 if (fp + tn) > 0 else 0

hw_metrics = self.profiler.profile_model(model, logic, X_train.shape,
device_name)

return {
    "Algorithm": algo_name,
    "Accuracy (%)": acc,
    "Precision (%)": prec,
    "Recall/Detection Rate (%)": rec,
    "F1-Score (%)": f1,
    "False Positive Rate (%)": fpr,
    "Training Time (ms)": (end - start) * 1000,
    **hw_metrics
}

```

Training_Lab.py

```

import streamlit as st
import pandas as pd
import os
from core.registry import CATEGORIES
from core.trainer import StudentTrainer
from core.profiler import EdgeProfiler
from core.config_user import COPYRIGHT_TEXT, PROJECT_TITLE

if 'results' not in st.session_state:
    st.session_state.results = []

st.set_page_config(page_title="Training", layout="wide")
st.title(f"🚀 Training Lab: {PROJECT_TITLE}")
st.sidebar.markdown("---")
st.sidebar.info(COPYRIGHT_TEXT)

profiler = EdgeProfiler()

with st.sidebar:
    st.header("1. Target Device")
    device = st.selectbox("Hardware", profiler.get_device_list())

    st.header("2. Dataset")

```

```

dataset = st.selectbox("Data", [f for f in os.listdir("data") if f.endswith(".csv")])

st.header("3. Algorithms")
all_models = []
# Prioritize Custom and Core
if "My Selected Algorithms" in CATEGORIES:
    all_models += CATEGORIES["My Selected Algorithms"]
all_models += CATEGORIES["Core Requirements"]
selected = st.multiselect("Select Algorithms", list(dict.fromkeys(all_models)))

st.header("4. Tuning")
hp_epochs = st.slider("Epochs (DL)", 1, 20, 5)
hp_batch = st.select_slider("Batch Size", [8, 16, 32, 64], value=32)
hp_est = st.slider("Estimators (RF)", 10, 100, 20)

if st.button("🚀 START TRAINING"):
    if not selected:
        st.error("Select algorithm.")
    else:
        trainer = StudentTrainer()
        st.session_state.results = []
        bar = st.progress(0)
        for i, algo in enumerate(selected):
            try:
                res = trainer.train_model(algo, f"data/{dataset}", device, hp_epochs,
hp_batch, hp_est)
                st.session_state.results.append(res)
            except Exception as e:
                st.error(f"Error ({algo}): {e}")
            bar.progress((i+1)/len(selected))
        st.success("Training Complete!")

if st.session_state.results:
    st.write(f"## Results for {device}")
    st.dataframe(pd.DataFrame(st.session_state.results))

```

Display_Page.py

```

import streamlit as st
import pandas as pd
import plotly.express as px
from core.config_user import COPYRIGHT_TEXT, PROJECT_TITLE

st.set_page_config(page_title="Display", layout="wide")
st.title(f"📊 Hypothesis Validation: {PROJECT_TITLE}")
st.sidebar.markdown("---")
st.sidebar.info(COPYRIGHT_TEXT)

if not st.session_state.results:
    st.warning("No results found. Train models first.")
    st.stop()

```

```

df = pd.DataFrame(st.session_state.results)

# HYPOTHESIS 2: Latency Threshold
st.subheader("🚀 Hypothesis 2: Edge AI Latency Analysis")
threshold = 25.0
df['Meets WiFi 7 Requirement'] = df['Latency (ms)'] < threshold

fig_lat = px.bar(df, x="Algorithm", y="Latency (ms)", color="Meets WiFi 7 Requirement",
                 color_discrete_map={True: "green", False: "red"},
                 text="Latency (ms)")
fig_lat.add_hline(y=threshold, line_dash="dash", line_color="red",
                 annotation_text="WiFi 7 Max Limit (25ms)")
st.plotly_chart(fig_lat, use_container_width=True)

# HYPOTHESIS 3: Detection Rate vs False Positive Rate
st.subheader("🔍 Hypothesis 3: False Positive Analysis")
fig_perf = px.scatter(df, x="False Positive Rate (%)", y="Recall/Detection Rate (%)",
                     size="Accuracy (%)", color="Algorithm",
                     title="Goal: High Detection Rate (Top) and Low FPR (Left)")
st.plotly_chart(fig_perf, use_container_width=True)

st.subheader("📄 Performance Summary Table")
st.dataframe(df)

```

Download_Page.py

```

import streamlit as st
import pandas as pd
import os
from core.packager import create_zip
from core.config_user import COPYRIGHT_TEXT, PROJECT_TITLE

st.set_page_config(page_title="Download", layout="wide")
st.title(f"📄 Download: {PROJECT_TITLE}")
st.sidebar.markdown("---")
st.sidebar.info(COPYRIGHT_TEXT)

if not st.session_state.results:
    st.warning("Please complete training first.")
    st.stop()

df = pd.DataFrame(st.session_state.results)

# Updated to match the new column name: 'F1-Score (%)'
best = df.loc[df['F1-Score (%)'].idxmax()]
st.success(f"🏆 Best Model based on F1-Score: **{best['Algorithm']}**")

```

```

col1, col2, col3 = st.columns(3)
col1.metric("Accuracy", f"{best['Accuracy (%)']:.2f}%")
col2.metric("Detection Rate", f"{best['Recall/Detection Rate (%)']:.2f}%")
col3.metric("FPR", f"{best['False Positive Rate (%)']:.4f}%")

st.markdown("---")
st.markdown("### 📦 Generate Deployment Firmware")
zip_path = create_zip(best['Algorithm'])
with open(zip_path, "rb") as f:
    st.download_button(
        label=f"⬇️ Download {best['Algorithm']} for Raspberry Pi",
        data=f,
        file_name=os.path.basename(zip_path),
        mime="application/zip"
    )

```

Appendix C: Github Repository Link

Link: <https://github.com/chnlhani227-max/FYP1>

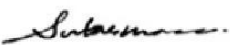


Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 3/11/2025	MEETING NO.: 1
PROJECT ID: T88J030	
PROJECT TITLE : Intrusion Detection in WiFi 7 IoT Networks using Edge AI	
TRIMESTER : Oct/Nov 2025	SUPERVISOR : Dr. Subarmaniam A/L Kannan
STUDENT ID & Name: 243UT2464K Hani Masyitah binti Hamzah	CO- SUPERVISOR : -

All to be filled in by student

1. WORK DONE - Project scope discussion.
2. WORK TO BE DONE - Install and set up Linux Mint as an environment.
3. PROBLEMS ENCOUNTERED - Not sure about the correct installation procedure.
4. COMMENTS


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 Multimedia University
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 75450 Melaka, Malaysia



 Supervisor's Signature &
 Stamp

 Co-Supervisor's Signature
 & Stamp (if any)

 Student's Signature

NOTES:

- Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
- For FYP Phase 1, total six log sheets are to be submitted (every other week*).
- For FYP Phase 2, total six log sheets are to be submitted (every other week**).
- Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

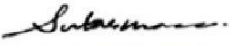


Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 6/11/2025	MEETING NO.: 2
PROJECT ID: T88J030	
PROJECT TITLE : Intrusion Detection in WiFi 7 IoT Networks using Edge AI	
TRIMESTER : Oct/Nov 2025	SUPERVISOR : Dr. Subarmaniam A/L Kannan
STUDENT ID & Name: 243UT2464K Hani Masyitah binti Hamzah	CO- SUPERVISOR : -

All to be filled in by student

1. WORK DONE - Linux Mint successfully installed. - Explored concepts and approaches to intrusion detection models.
2. WORK TO BE DONE - Identify the specific Edge AI frameworks and WiFi 7 simulation tools to be used. - Create system design (diagrams)
3. PROBLEMS ENCOUNTERED
4. COMMENTS


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Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature
-----------------------------------	---	---------------------

NOTES:

5. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
6. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
7. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
8. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

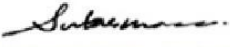


Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 23/11/2025	MEETING NO.: 3
PROJECT ID: T88J030	
PROJECT TITLE : Intrusion Detection in WiFi 7 IoT Networks using Edge AI	
TRIMESTER : Oct/Nov 2025	SUPERVISOR : Dr. Subarmaniam A/L Kannan
STUDENT ID & Name: 243UT2464K Hani Masyitah binti Hamzah	CO- SUPERVISOR : -

All to be filled in by student

1. WORK DONE - Library Literacy Course (LLC) meeting - FYP For Fist Class 3/2025 - 11th December 2025 Thursday 10:00am – 12:00pm
2. WORK TO BE DONE
3. PROBLEMS ENCOUNTERED
4. COMMENTS


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Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature
-----------------------------------	---	---------------------

NOTES:

9. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
10. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
11. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
12. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

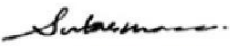


Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 11/12/2025	MEETING NO.: 4
PROJECT ID: T88J030	
PROJECT TITLE : Intrusion Detection in WiFi 7 IoT Networks using Edge AI	
TRIMESTER : Oct/Nov 2025	SUPERVISOR : Dr. Subarmaniam A/L Kannan
STUDENT ID & Name: 243UT2464K Hani Masyitah binti Hamzah	CO- SUPERVISOR : -

All to be filled in by student

1. WORK DONE - Created 13 types of diagrams for system design.
2. WORK TO BE DONE - Find related papers for Literature Review.
3. PROBLEMS ENCOUNTERED
4. COMMENTS


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 Supervisor's Signature &
 Stamp

 Co-Supervisor's Signature
 & Stamp (if any)

 Student's Signature

NOTES:

13. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
14. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
15. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
16. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

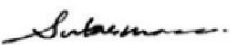


Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 15/12/2025	MEETING NO.: 5
PROJECT ID: T88J030	
PROJECT TITLE : Intrusion Detection in WiFi 7 IoT Networks using Edge AI	
TRIMESTER : Oct/Nov 2025	SUPERVISOR : Dr. Subarmaniam A/L Kannan
STUDENT ID & Name: 243UT2464K Hani Masyitah binti Hamzah	CO- SUPERVISOR : -

All to be filled in by student

1. WORK DONE - Extracted and analysed 41 papers for Literature Review. - Identify research gaps.
2. WORK TO BE DONE - Begin writing the final Literature Review.
3. PROBLEMS ENCOUNTERED
4. COMMENTS


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 75450 Melaka, Malaysia



 Supervisor's Signature &
 Stamp

 Co-Supervisor's Signature
 & Stamp (if any)

 Student's Signature

NOTES:

17. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
18. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
19. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
20. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

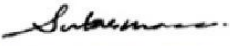


Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 9/1/2025	MEETING NO.: 6
PROJECT ID: T88J030	
PROJECT TITLE : Intrusion Detection in WiFi 7 IoT Networks using Edge AI	
TRIMESTER : Oct/Nov 2025	SUPERVISOR : Dr. Subarmaniam A/L Kannan
STUDENT ID & Name: 243UT2464K Hani Masyitah binti Hamzah	CO- SUPERVISOR : -

All to be filled in by student

1. WORK DONE - Introduction, Literature Review, Methodology, and Research Design have been drafted. - Simulation testing is running.
2. WORK TO BE DONE - Fix simulation code. - Replace smaller datasets used for early testing with the larger, more comprehensive datasets to ensure the simulation's validity.
3. PROBLEMS ENCOUNTERED - Datasets initially downloaded and used for early-stage simulation were smaller than required.
4. COMMENTS


 DR. SUBARMANIAM A/L KANNAN
 Asst. Professor
 Faculty of Information Science and Technology
 Multimedia University
 Jalan Ayer Keroh Lama
 75450 Melaka, Malaysia



Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature
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NOTES:

21. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
22. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
23. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
24. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)